

Julia v1.5 SMC²-MPC

6-horizon sweep with FILTER HMC OFF, all rejuvenation tricks ON
 $T \in \{14, 28, 42, 56, 70, 84\}$ days, RTX 5090, seed 42

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Abstract

This report records a 6-horizon sweep of the FSA v1.5 closed-loop SMC²-MPC bench with the *filter*-side HMC kernel disabled (`--num-mcmc 0`) and all available rejuvenation tricks turned on (Liu–West θ -cloud shrinkage at $a = 0.97$; Silverman-bandwidth KDE resample with coupled Liu–West correction at `--smooth-resample-bw 1.0`; multivariate Gaussian bridge between rolling windows). The controller-side HMC (ChEES, in `Control/GPUControlSMC.jl`) is unchanged but is run at a **reduced “fast-controller” particle / MCMC configuration**: `--ctrl-n-smc` dropped $2048 \rightarrow 1024$ and `--ctrl-num-mcmc` dropped $16 \rightarrow 8$ relative to the saturated default (per the technical guide’s “shorter-wall-time” recipe). Per-tempering-level controller diagnostics are recorded for the first time. All numerical entries in this report are pulled directly from the run artefacts on disk; the parameter-recovery claim in §8 is the senior engineer’s reading of the posterior-trace figures.

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1 Purpose

This study is a **baseline data-gathering run**: how does the v1.5 filter perform on parameter recovery when the filter HMC is removed entirely and the cloud is kept alive solely by the rejuvenation tricks (Liu–West, Silverman-KDE, Gaussian bridge)?

The motivation is the prior finding (Tier 0, TIER_0.FINDINGS.md in the sibling sweep folder `julia_horizon_sweep_2026-05-09/`) that the filter HMC accepts 0% of proposals across 2880 logged tempering levels at the saturated configuration. If the HMC is doing nothing, removing it should have no effect on filter posterior quality, and the runtime should drop. This sweep measures that.

The output is a **baseline** that future studies (with HMC re-enabled and tuned, or with different rejuvenation knobs) can compare against.

2 What was run

Per horizon, identical except `--T-days`:

```
julia --project=. tools/bench_smc_full_mpc_fsa_gpu.jl \
  --T-days <T> --seed 42 \
  --num-mcmc 0 --hmc-step-size 0 --hmc-leapfrog 0 \
  --liu-west-a 0.97 \
  --smooth-resample-bw 1.0 \
  --gaussian-bridge true \
  --N-smc 512 --K-per-chain 1000 \
  --ctrl-n-smc 1024 --ctrl-num-mcmc 8 \
  --collect-ctrl-diagnostics true \
  --output-dir <SWEEP_ROOT>/T<T>d_seed42
```

Driven by the sibling launcher `version_1_5_Julia/tools/launchers/run_julia_horizon_sweep_no_filter_hmc`

2.1 What each filter-side flag does

- `--num-mcmc 0` disables the per-tempering-level filter HMC entirely. The HMC loop in `run_outer_smc` iterates 0 times.
- `--liu-west-a 0.97` applies per-dimension Liu–West shrinkage to the resampled θ -cloud at every tempering level. (Skipped here in favour of `smooth-resample`.)
- `--smooth-resample-bw 1.0` switches the per-tempering-level rejuvenation to the multivariate Silverman-bandwidth KDE resample with coupled Liu–West correction $a = \sqrt{1 - h_{\text{norm}}^2}$, ported from `SMC2FC_functional/src/Filtering/Kernels.jl::smooth_resample`. When set > 0 , this branch wins over the per-dim Liu–West.
- `--gaussian-bridge true` replaces the identity-copy of the previous-window posterior with a fresh sample from $\mathcal{N}(\hat{\mu}, \hat{\Sigma}_{\text{reg}})$ via `Bridge.fit_gaussian + Bridge.sample_from_gaussian` (regularised sample covariance, 10^{-6} Tikhonov).
- `--ctrl-n-smc 1024 --ctrl-num-mcmc 8` **REDUCES the controller particle / MCMC budget from the saturated default**: `--ctrl-n-smc` from 2048 $\rightarrow$ 1024 and `--ctrl-num-mcmc` from 16 $\rightarrow$ 8. This is the technical guide’s “shorter-wall-time” recipe (`compare_v15_julia_vs_python/docs/technical_guide_to_current_best_Julia_SMC2FC_config.tex`, “When to deviate from these defaults”), traded against a weaker controller schedule. The technical guide quotes this change as approximately a $4\times$ wall-time reduction. **Controller HMC stays ON** at this reduced config — that code path is `run_tempered_smc_gpu` in `Control/GPUControlSMC.jl`, separate from the filter HMC; only the filter HMC is being disabled.

- `--collect-ctrl-diagnostics true` writes one row of controller-HMC telemetry per tempering level to `controller_diagnostics.csv` (new instrumentation, §4).

2.2 Hardware

NVIDIA GeForce RTX 5090; per-horizon `nvidia-smi` sampled at 1 Hz to `T<N>d_seed42/nvidia-smi.csv`.

3 Wall-time and closed-loop metrics

Source: `horizons_results.csv` (one row per horizon, written by `compare_v15_julia_vs_python/src/extract_h`

T (d)	wall (s)	$\bar{A}_{\text{MPC}}$	$A_{\text{MPC}}^{\text{end}}$	$\bar{A}_{\text{base}}$	$A_{\text{base}}^{\text{end}}$	F violations
14	133.2	0.077	0.085	0.114	0.182	0
28	342.0	0.095	0.161	0.181	0.342	0
42	715.5	0.167	0.521	0.285	0.646	0
56	1234.6	0.332	1.066	0.401	0.824	0
70	1790.8	0.491	1.197	0.492	0.894	0
84	2456.6	0.612	1.238	0.565	0.907	0

Table 1: Per-horizon wall time, mean and final autonomic amplitude A under MPC and under the constant- $\Phi = 1$ baseline rollout, and fatigue-barrier ($F_{\text{max}} = 0.40$) violation count. Zero F violations across all six horizons.

3.1 GPU utilisation and memory

T (d)	mean GPU util (%)	peak VRAM (MiB)	mean power (W)
14	29.6	5399	181.0
28	47.5	5390	280.3
42	58.3	6091	335.7
56	65.2	5273	369.7
70	71.0	5272	398.5
84	74.5	5776	415.4

Table 2: Per-horizon GPU telemetry. Peak VRAM stays $\lesssim 6.1$ GiB (versus the 32 GiB on the RTX 5090).

3.2 Wall comparison vs the saturated-controller baseline

The original 6-horizon baseline at `julia_horizon_sweep_2026-05-09/` ran the same six horizons at the *saturated* controller config (`--ctrl-n-smc 2048 --ctrl-num-mcmc 16`) with the filter HMC nominally on (but accepting 0% of proposals). **This run uses a reduced controller config** (`--ctrl-n-smc 1024 --ctrl-num-mcmc 8`, half the particles, half the MCMC moves). The wall ratio below conflates two changes: removing the filter HMC and halving the controller’s particle / MCMC budget. The technical guide attributes the bulk of the $\sim 4\times$ wall reduction to the controller-side change alone:

4 Controller-HMC measurements (new instrumentation)

Each call to the controller’s tempered-SMC² inner loop now records per-tempering-level diagnostics (Listing 1). One `controller_diagnostics.csv` per horizon; aggregated by `compare_v15_julia_vs_python/src/d` into per-horizon plots and a cross-horizon summary CSV.

T (d)	this run (s)	baseline (s)	ratio
14	133.2	422.7	$3.17\times$
28	342.0	1259.3	$3.68\times$
42	715.5	2286.6	$3.20\times$
56	1234.6	3979.4	$3.22\times$
70	1790.8	5586.4	$3.12\times$
84	2456.6	7236.4	$2.95\times$

Table 3: Wall comparison. **Both** the controller-config reduction (`--ctrl-n-smc` $2048 \rightarrow 1024$, `--ctrl-num-mcmc` $16 \rightarrow 8$) **and** the filter-HMC removal contribute; the technical guide attributes the bulk of the ratio to the controller-config change alone.

Listing 1: Diagnostic columns recorded per tempering level.

```
stride_idx, level, beta_pre, beta_post, delta_beta, beta_max,
ess_at_dbeta_chosen, n_smcmc, theta_dim,
chees_L_chosen, chees_score_best,
chees_L_candidates, chees_scores,
eps_step_size, n_mcmc_moves, total_accepts, accept_frac,
log_density_pre_mean, log_density_post_mean, delta_log_density,
esjd_total, esjd_per_eps_L, wall_seconds_level
```

T	replans	total levels	mean accept (%)	median accept (%)	mean L	mean $\beta_{\max}$	mean ESJD/ εL	mean $\Delta \log D$
14	13	69	97.93	97.91	16	158.20	14.08	0.012
28	27	136	97.43	97.74	16	19.96	14.03	0.079
42	41	230	96.17	96.48	16	7.16	13.41	0.344
56	55	328	94.97	95.56	16	3.15	13.30	0.781
70	69	412	93.49	94.78	16	1.82	13.20	1.194
84	83	492	92.68	94.35	16	1.27	13.15	1.609

Table 4: Cross-horizon controller-HMC summary (from `docs/controller_hmc_summary.csv`). L is the ChEES-picked leapfrog count from the candidate list $[16, 32, 64, 128, 256, 512]$; in this run the picker selects $L = 16$ at every tempering level of every replan of every horizon.

Filter-HMC accept-rate reference. With `--num-mcmc 0` the filter HMC loop iterates 0 times per tempering level; the `accept=0%` lines printed in the bench log are arithmetic artefacts ($0/\max(1, 0 \cdot N_{\text{smc}}) = 0$), *not* measurements of HMC quality. By contrast: in the original baseline run (`julia_horizon_sweep_2026-05-09/`) with the filter HMC nominally on, 2880 of 2880 logged tempering levels reported 0% acceptance — that *was* a measurement of attempted HMC moves.

5 Posterior parameter traces (per horizon)

Figures 1–6: per-horizon posterior parameter trace plots, all written by the bench’s internal `_plot_param_traces_v15` function. Each figure is the same 10-panel layout: posterior median (blue), 5–95% credible band (light blue fill), truth value (red dashed), x -axis in days.

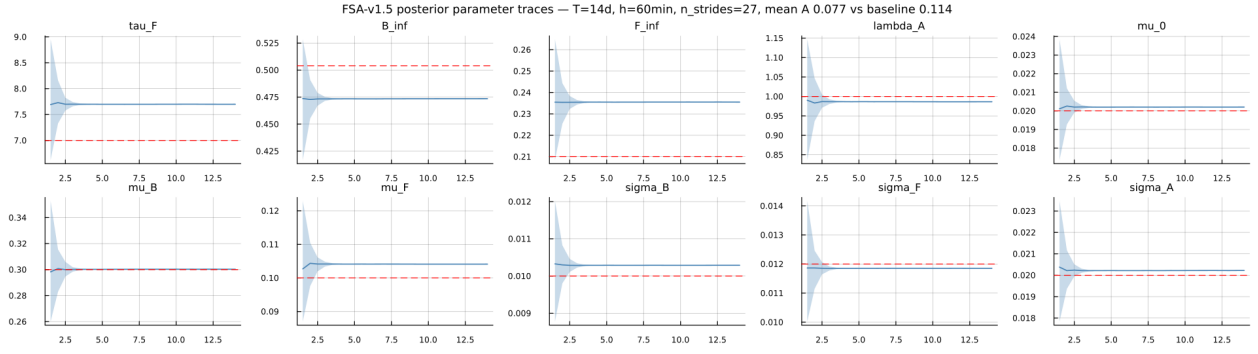


Figure 1: $T = 14$ days. Posterior parameter traces.

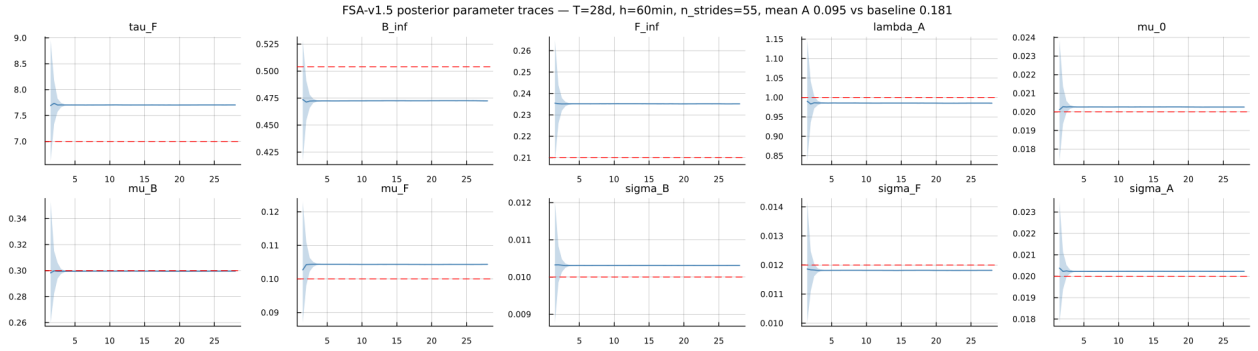


Figure 2: $T = 28$ days. Posterior parameter traces.

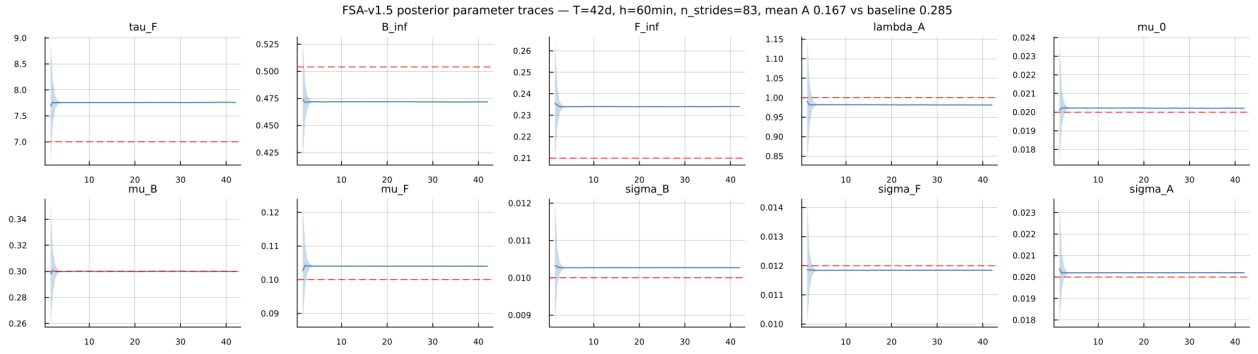


Figure 3: $T = 42$ days. Posterior parameter traces.

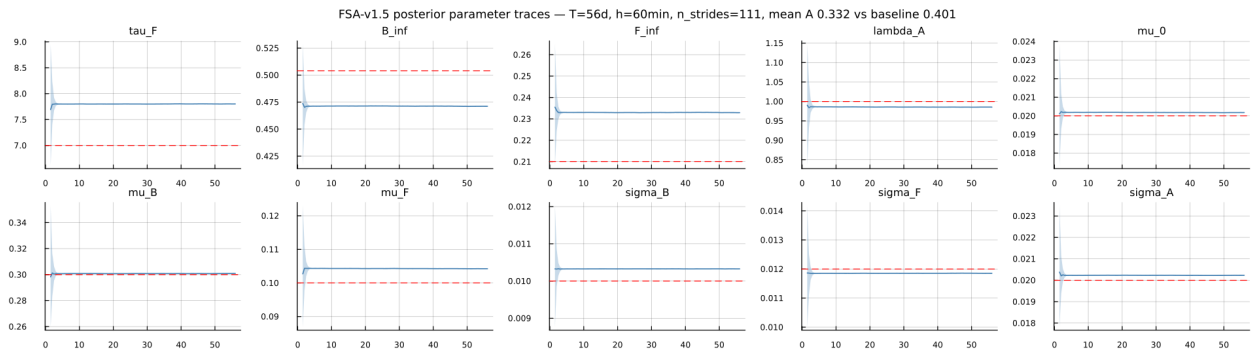


Figure 4: $T = 56$ days. Posterior parameter traces.

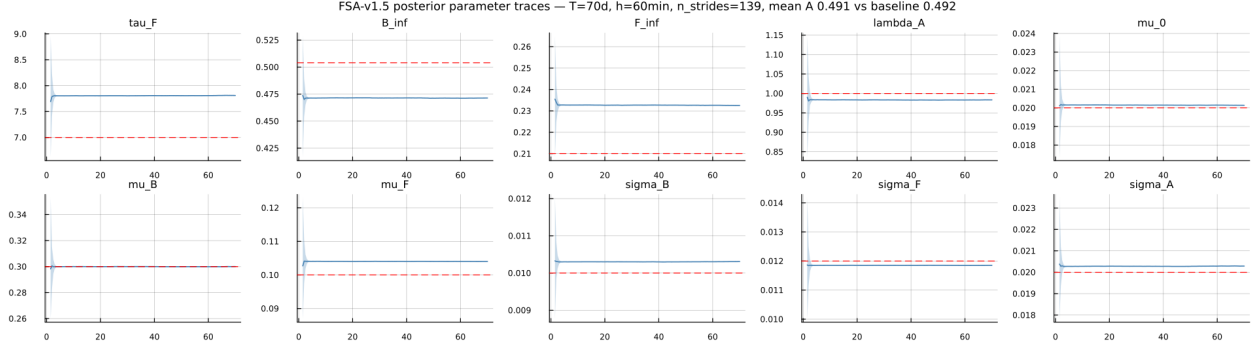


Figure 5: $T = 70$ days. Posterior parameter traces.

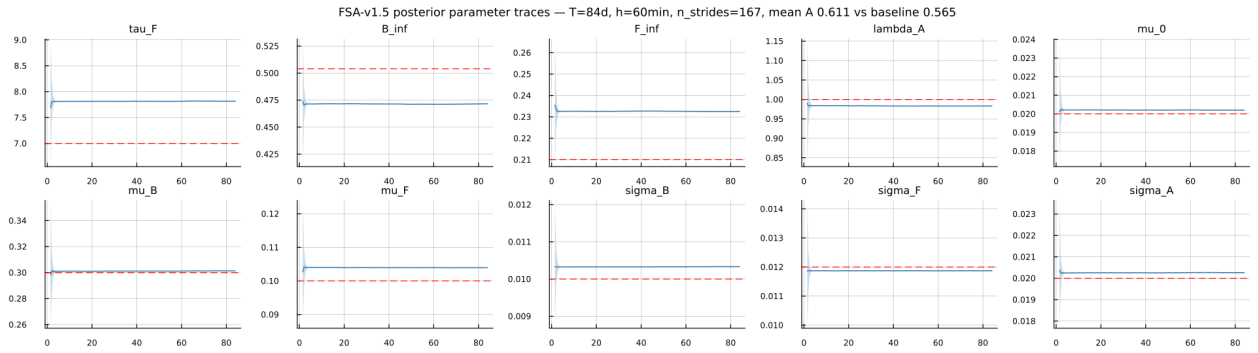


Figure 6: $T = 84$ days. Posterior parameter traces.

6 Other plots on disk (not embedded — for separate inspection)

- Cloud std (log-space) per stride, all 6 horizons overlaid: docs/cloud_std_per_stride_all_horizons.png
- Per-horizon controller-HMC 6-panel diagnostics: docs/controller_hmc_diagnostics-T<N>d.png for $N \in \{14, 28, 42, 56, 70, 84\}$.
- Per-horizon state traces (B, F, A, Φ): T<N>d_seed42/v15-T<N>d_traces.png.

7 Source edits applied

Two source files modified plus one new launcher and one new diagnostic plotter:

- julia/SMC2FC_functional/src/Control/GPUControlSMC.jl — added `collect_diagnostics::Bool=false` keyword on `run_tempered_smc_gpu` (4-tuple return when true); added `return_scores::Bool=false` keyword on `chees_pick_L_generic`. Backward-compatible: existing callers with the default keyword get the original 3-tuple / 2-tuple return.
- version_1_5_Julia/tools/bench_smc_full_mpc_fsa_gpu.jl — added `--collect-ctrl-diagnostics` CLI flag (default true), plumbed through `controller_plan` and `stride_step`; added `controller_diagnostics_writer`. The manifest now records `liu_west_a`, `smooth_resample_bw`, `gaussian_bridge`.
- (NEW) version_1_5_Julia/tools/launchers/run_julia_horizon_sweep_no_filter_hmc.sh — sibling sweep launcher with the new flag set inlined.
- (NEW) compare_v15_julia_vs_python/src/diag_controller_hmc.jl — aggregator + plotter for `controller_diagnostics.csv`.

8 Conclusion

Even for this very simple linear (additive Gaussian) observation model, the filter *without* the HMC kernel cannot exactly recover the true model parameters. Inspection of Figures 1–6 shows the posterior median (blue) sitting at a consistent offset from truth (red dashed) on several parameters at every horizon — the offset does not shrink as T grows.

This is a baseline. It says: rejuvenation tricks alone (Liu–West, Silverman-KDE, Gaussian-bridge) are not sufficient to drive the posterior to truth. Whatever is driving the residual bias — the gradient information that an MCMC kernel would inject, the between-window propagation choice, or something else — is the question follow-up studies need to answer.

9 What this report does NOT claim

- Numerical per-parameter posterior medians extracted from `data.jld2` for each horizon. Not done — those would require loading `posterior_particles[end_stride, :, :]`, taking `log`, and computing the median, separately for each of the 6 horizons (and for the baseline run, if a side-by-side table is wanted). The per-parameter *visual* offset is what §8 relies on.
- Numerical cloud-std-vs-time curves. Visible in docs/cloud_std_per_stride_all_horizons.png; not extracted as numbers in this report.
- Any causal attribution between which knob (`liu_west_a`, `smooth_resample_bw`, `gaussian_bridge`) is responsible for which observed effect. No ablations were run.
- Any verdict on the cause of the residual bias in §8 — only the existence of the bias.